

Research Interests

Statistical learning for complex temporal and functional data, with a focus on functional data analysis and causal inference.

Education

Emory University

Ph.D. in Biostatistics

Advisor: Prof. Julia Wrobel

Atlanta, US
Expected June 2029

Beijing University of Technology (BJUT)

B.S. in Statistics; Minor in Computer Science and Technology

GPA: 3.95/4.0; Class Rank: 1/26

BJUT First-class Scholarship for four consecutive years

Beijing, China
June 2024

Research Experience

Functional Registration and Dimension Reduction for Binary Activity Curves

Ph.D. Student, advised by Prof. Julia Wrobel

Atlanta, US
June 2025–Present

- Developed an MM-based registration framework for binary functional observations, estimating subject-specific inverse warping functions under monotonicity constraints.
- Implemented a variational EM-based FPCA module to recover latent probability curves and subject-specific functional random effects from binary longitudinal data.
- Built a scalable R/Rcpp pipeline that alternates between template estimation and time-warping updates for large-scale minute-level accelerometer data.
- Evaluated the method on simulations and NHANES activity data, including complete-case preprocessing, covariate-adjusted age analyses, and visualization of aligned activity patterns.

Sparse Change-Point Analysis of Eye Movement and Mind Wandering

Researcher and Co-author, supervised by Dr. Hui Lan

Beijing, China
May 2023–June 2024

- Processed high-dimensional sparse experimental data from 300 participants, including 23 behavioral and facial-expression variables, using R.
- Reproduced high-dimensional multivariate change-point detection methods, including SARA and SBS, as benchmark models.
- Proposed a sparsity-aware CUSUM statistic and compared it with existing methods; contributed to statistical analysis and manuscript preparation.

YOLOX-Based Livestock Object Detection in Remote Sensing Imagery

Researcher, supervised by Dr. Qi Wang, Chinese Academy of Sciences

Hulun Buir, China
July 2023–April 2024

- Collected aerial livestock imagery in Hulun Buir using UAVs and tethered balloons, producing 1,245 images and over 90 hours of video.
- Created 628 labeled training images through manual annotation and trained an improved YOLOX-based object detection model.
- Improved detection accuracy from 65% to 73% through data augmentation and a binary CNN-based flock classification module; evaluated models using accuracy, precision, and recall.

Generalized Least Squares Calibration for Expensive Computer Models

Researcher and Co-author, supervised by Dr. Yan Wang

Beijing, China
Oct. 2022–Mar. 2024

- Built Gaussian process surrogate models to approximate outputs from expensive computer experiments.
- Derived a variance-augmented loss formulation to stabilize calibration and improve predictive accuracy.
- Reproduced KO and GALS calibration methods using data generated from the MATLAB FEM toolbox.

Internship Experience

ByteDance

Datacenter Operations Engineer Intern, Global Monetization - Business Integrity

Beijing, China
Jan. 2024–Mar. 2024

- Supported the global rollout of the Workforce Management (WFM) platform for ByteDance teams across the US and Asia-Pacific regions.
- Developed and standardized quantitative performance metrics, including Volume and Manpower Matching Degree (VMMD) and Cost Per Ads (CPA), replacing the prior CPST indicator.

Guorong Securities

Intern, Business Management Division, supervised by Ms. Yu Wu

Beijing, China
July 2022–Aug. 2022

- Analyzed recruitment seasonality and labor-market patterns through data processing, visualization, and statistical modeling.
- Applied time-series and NLP models, including ARIMA, SARIMA, VARIMA, LSTM, and XLNet, to improve extraction accuracy and support timed advertisement placement.
- Proposed data-driven solutions addressing data gaps, competitor activity, COVID-19 disruptions, and industry dynamics.

Academic Projects

Quantitative Modeling of Species Competition under Irregular Weather Cycles

Team Leader, supervised by Prof. Min Han

Feb. 2023–Mar. 2023

- Modeled species interactions under irregular weather cycles using niche difference, competitive difference, and Lotka–Volterra competition models.
- Simulated precipitation patterns with a random-walk algorithm and analyzed individual-level variation under changing environmental conditions.

Regression Analysis of Luckin Coffee Purchase Behavior at BJUT

Researcher, supervised by Dr. Yan Wang

Sept. 2022–Dec. 2022

- Designed an online survey, completed data screening and preprocessing, and conducted descriptive analysis and visualization in R.
- Compared exponential, linear, quadratic, and segmented regression models and performed model diagnostics using AIC.

Epidemic Modeling with SIR, SEIR, and SEINR Models

Researcher, supervised by Prof. Xinyuan Zhao

Aug. 2022–Sept. 2022

- Processed official epidemic datasets and simulated outbreak dynamics under SIR, SEIR, and SEINR frameworks.
- Incorporated inter-city migration and partitioned data into training, testing, and validation sets for model comparison.

Skills

Programming and Tools: R, Rcpp, Python, C/C++, SQL, MATLAB, PyTorch, TensorFlow, SPSS, Excel

Statistical Methods: Functional data analysis, FPCA, functional registration, causal inference, time-series modeling, anomaly detection, regression, Gaussian processes

Languages: Mandarin Chinese (native), English (fluent)